Technical specification template: Low-Voltage Switchgear and Assemblies

OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. Package revision A5. Generated 2026-11-05 07:50 UTC. Package version 0.1.0.dev0.

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Document control

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1. Scope

This template defines the technical requirements for low-voltage switchgear and assemblies for the OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. It is to be completed and issued by the project. Items in [square brackets] are decisions or data the project must supply. Preliminary values from the Ohmetry reference configuration are shown with their data status and must be replaced or confirmed.

2. Applicable standards

- IEC 61439-1 and IEC 61439-2 (low-voltage switchgear and controlgear assemblies), design verification
- IEC 60947-2 (circuit-breakers)
- IEC 61869 series (instrument transformers)
- Local regulations [project to add]
- Editions and applicability of every standard above are NOT VERIFIED. The project shall confirm the current editions and any national deviations.

3. Service conditions and system data

Item	Requirement or preliminary value	Data status
Nominal LV system	400 V line-to-line, 60 Hz, 3-phase	PRELIMINARY_ASSUMPTION. Utility confirmation is the owner's statement, not independently verified
MV supply	13.8 kV	PRELIMINARY_ASSUMPTION. To be confirmed with the utility
Design ambient temperature	30 degC	PRELIMINARY_ASSUMPTION. Site data to be confirmed
Site altitude	0 m	PRELIMINARY_ASSUMPTION. Site data to be confirmed
Prospective fault level	[To be calculated: short-circuit	NOT CALCULATED

	study pending utility, transformer and generator data]	

4. Equipment ratings

Ratings below are preliminary values from the master configuration unless marked otherwise. They are not manufacturer-confirmed and are not an equipment selection.

Item	Requirement or preliminary value	Data status and note
Rated voltage and frequency	400 V, 60 Hz	PRELIMINARY_ASSUMPTION
Busbar rated current	[At least the incomer design current after derating; preliminary 4,136 A (path-out)]	Preliminary requirement, pending manufacturer data
Rated short-time withstand (kA, 1 s) and peak	[To be at least the calculated LV fault level]	NOT CALCULATED: utility, transformer impedance and generator data pending
Form of separation, IP rating	[Project to select]	Project decision
Circuit-breaker breaking and making capacity	[At least the fault level at each location]	Pending manufacturer data

5. Technical requirements

5.1 The assembly shall be design verified per IEC 61439-1 and -2; the verification evidence (test, calculation or design rule) shall be supplied for each rating, with the assembly manufacturer identified.

5.2 The short-circuit withstand shall exceed the prospective fault level including the peak, at the busbars and at each outgoing device.

5.3 Circuit-breakers shall have breaking capacity above the fault level at their location; time-current characteristics shall be supplied for the coordination study.

5.4 Temperature rise at the rated current and the site ambient shall be evidenced; derating of busbars at altitude and ambient shall be stated.

5.5 Metering, monitoring, interlocking, transfer scheme and arc-fault mitigation: [project to define].

6. Manufacturer data required

The tenderer shall supply the datasheets and values below with its offer, each with the document name and revision. A value without a source document is not accepted. These are also tracked in the manufacturer-data register and are imported into the calculations through the data override mechanism.

Parameter id	Parameter	Datasheet	Current status
distribution.LV-MSB-A.rated_current_a	LV-MSB-A: rated_current_a	LV switchgear data sheet and IEC 61439 verification evidence	MISSING
distribution.LV-MSB-A.short_circuit_rating_ka	LV-MSB-A: short_circuit_rating_ka	LV switchgear data sheet and IEC 61439 verification evidence	MISSING
distribution.LV-MSB-A.form_of_separation	LV-MSB-A: form_of_separation	LV switchgear data sheet and IEC 61439	MISSING

		verification evidence	
distribution.LV-MSB-A.ip_rating	LV-MSB-A: ip_rating	LV switchgear data sheet and IEC 61439 verification evidence	MISSING
distribution.LV-MSB-A.verification_evidence	LV-MSB-A: verification_evidence	LV switchgear data sheet and IEC 61439 verification evidence	MISSING
distribution.ACB-TX-A.breaking_capacity_ka	ACB-TX-A: breaking_capacity_ka	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.ACB-TX-A.making_capacity_ka	ACB-TX-A: making_capacity_ka	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.ACB-TX-A.time_current_curve_ref	ACB-TX-A: time_current_curve_ref	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.ACB-TX-A.settings	ACB-TX-A: settings	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.ACB-GEN-A.breaking_capacity_ka	ACB-GEN-A: breaking_capacity_ka	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.ACB-GEN-A.making_capacity_ka	ACB-GEN-A: making_capacity_ka	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.ACB-GEN-A.time_current_curve_ref	ACB-GEN-A: time_current_curve_ref	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.ACB-GEN-A.settings	ACB-GEN-A: settings	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.ACB-UPS-IN-A.breaking_capacity_ka	ACB-UPS-IN-A: breaking_capacity_ka	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.ACB-UPS-IN-A.making_capacity_ka	ACB-UPS-IN-A: making_capacity_ka	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.ACB-UPS-IN-A.time_current_curve_ref	ACB-UPS-IN-A: time_current_curve_ref	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING
distribution.ACB-UPS-IN-A.settings	ACB-UPS-IN-A: settings	Circuit breaker or relay data sheet with time-current curves and test certificates	MISSING

7. Inspection and quality requirements

7.1 Review of design verification evidence, drawings and the ITP.

7.2 Inspection of busbar supports, clearances, wiring and labelling.

7.3 Routine tests on every unit as required by the applicable standard and this specification.

7.4 Type-test reports or evidence for the design, issued by an accredited or recognised laboratory [project to define acceptable evidence].

7.5 Inspection and test plan (ITP) approved before manufacture, naming witness and hold points.

7.6 Calibration certificates for test instruments.

8. Factory acceptance test (FAT) requirements

8.1 Routine verification per IEC 61439: wiring, operation, dielectric, protective circuit continuity, mechanical function.

8.2 Breaker functional tests and settings checks; transfer scheme simulation.

8.3 Witnessing: [owner / independent inspector to attend].

8.4 A FAT procedure shall be submitted for approval before the test; the results shall be issued as a signed report, with deviations and their disposition recorded.

9. Commissioning requirements

9.1 Torque and insulation resistance, phase sequence, protection tests, interlock and transfer tests.

9.2 Load and thermography checks [project to define].

9.3 Commissioning records shall name the test, the instrument and its calibration, the date and the person, and shall be retained with the verification register.

10. Submittals and documentation

- Datasheets, drawings and type-test reports.
- Inspection and test plan, FAT procedure and report.
- Installation, operation and maintenance manuals.
- Spare parts and special tools lists.
- Warranty terms [project to define].

11. Open items (NOT VERIFIED)

- All ratings in section 4 marked PRELIMINARY or UNVERIFIED are open until confirmed from manufacturer data and the project studies.
- The prospective fault level is NOT CALCULATED until the utility, transformer and generator data are supplied.
- Standards editions and local requirements are NOT VERIFIED.
- This template has not been reviewed by an independent engineer.